

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour

Total: Marks:40

Annexure A

Code of Conduct:

I Shaik Fahed (192565026)___(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SHAIK FAHED

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diabetes Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them - justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naive Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Action Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context	
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts	
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution	
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools	
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis	
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation	

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive learning approach suitable for the medical dataset

The given healthcare dataset contains patient attributes such as age, blood pressure, cholesterol, glucose level, BMI and family history. A suitable approach is **supervised inductive learning**, because historical patient records can be used as labelled examples to learn a general rule for predicting diabetes in a new patient. A Decision Tree classifier can learn human-readable rules, while Logistic Regression can provide a probability of diabetes.

Target variable: Diabetes diagnosis — Diabetic (1) or Non-diabetic (0).

Key features and justification

- **Glucose level:** It is a major indicator of diabetes risk and can strongly separate diabetic and non-diabetic patients.
- **BMI:** BMI gives information about body weight relative to height and can contribute to diabetes risk.
- **Age:** Diabetes risk can vary with age, making age useful for risk stratification.
- **Blood pressure:** Abnormal blood pressure can provide additional information about a patient's metabolic and cardiovascular risk.
- **Cholesterol:** Cholesterol can provide supporting information about the patient's overall health profile.
- **Family history:** It represents inherited or familial risk and may improve prediction.

(b) Handling class imbalance

The dataset contains more non-diabetic than diabetic cases. If a classifier is trained directly on this imbalanced data, it may favour the majority class and miss diabetic patients. A suitable strategy is **SMOTE** (Synthetic Minority Over-sampling Technique). SMOTE creates synthetic minority-class examples from existing diabetic records rather than simply duplicating the same records.

- It increases representation of the diabetic class.
- It helps the model learn decision boundaries for minority cases.
- It can improve recall for diabetes detection.
- SMOTE should be applied only to the training portion after the train/test split to avoid data leakage.

Alternative: Class weights can assign a larger penalty to errors on diabetic cases. Under-sampling is possible, but it may discard useful non-diabetic information.

TASK 1 (CONTINUED): SPLITTING & PREPROCESSING

(c) 70:30 split and justification for a medical use-case

The dataset should be divided into **70% training data and 30% testing data**. The training set is used to learn the model, while the testing set remains unseen until final evaluation. The 70:30 ratio provides a sufficiently large learning sample while retaining enough unseen records for an independent performance estimate. A **stratified split** should be used so that the proportion of diabetic and non-diabetic patients is maintained in both sets.

Preprocessing steps

Step	Procedure	Impact
Missing values	Identify missing patient values and impute them using an appropriate statistical method or remove missing records.	Prevents statistical methods from crashing due to missing data.
Normalization	Scale numerical variables such as age, blood pressure, cholesterol, glucose and BMI to a comparable range.	Makes numerical features comparable and is particularly useful for L2 regularization.
Encoding	Convert categorical information such as family history into numerical form.	Allows machine-learning algorithms to process categorical variables.
Class balancing	Apply SMOTE or class weights to training data only.	Improves learning of the minority diabetic class.
Data leakage control	Fit preprocessing transformations using training data and then apply them to test data.	Prevents test data leakage.

Expected impact: Proper preprocessing improves model consistency, prevents numerical scale from affecting statistical learning unnecessarily, and helps the model detect diabetic cases without allowing information from the test set to influence training.

Important limitation: The assessment paper gives the patient attributes and tasks, but does not supply the actual patient records. Therefore, exact dataset-dependent numerical results must be obtained by training the models on the provided dataset, if one is supplied separately.

TASK 2: DECISION TREE FOR DIABETES DIAGNOSIS

(a) First three levels of the Decision Tree

A Decision Tree repeatedly selects a feature and threshold that best separates diabetic and non-diabetic patients. A reasonable illustrative structure is:

Level 0: Glucose level → Low/Normal or High

Level 1: Each branch can be divided using BMI

Level 2: Remaining branches can be divided using Age, Blood Pressure or Family History.

TASK 2: DECISION TREE FOR DIABETES DIAGNOSIS

Level	Illustrative decision	Purpose
0 – Root	Glucose $\leq$ threshold?	Separates patients according to an important diabetes-risk indicator.
1 – Left branch	BMI $\leq$ threshold?	Separates lower and higher BMI patients among lower-glucose cases.
1 – Right branch	BMI $\leq$ threshold?	Separates risk among higher-glucose cases.
2	Age / Blood Pressure / Family History	Further refines patient groups toward diabetic or non-diabetic leaves.

Root node selection using Information Gain

Information Gain measures how much uncertainty is reduced after splitting the data. It is calculated as:

$$\text{IG} = \text{Entropy}(\text{parent}) - \text{Weighted Entropy}(\text{children})$$

The feature with the highest Information Gain is selected as the split because it produces the greatest reduction in class uncertainty.

Root node selection using Gini Index

$$\text{Gini} = 1 - \sum p_i^2$$

The best split is the one that produces the largest reduction in Gini impurity. Glucose is a suitable candidate root because it is expected to separate diabetes-risk groups strongly; however, the exact root must be selected from the actual dataset using the calculated Information Gain/Gini values.

(b) Model evaluation

Metric	Formula	Interpretation
Accuracy	$(\text{TP} + \text{TN}) / \text{Total}$	Percentage of all predictions that are correct.
Precision	$\text{TP} / (\text{TP} + \text{FP})$	Of the patients predicted diabetic, how many are actually diabetic.
Recall	$\text{TP} / (\text{TP} + \text{FN})$	Of the actual diabetic patients, how many are detected.
F1-Score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Balances precision and recall.

False Negative: A false negative occurs when an actual diabetic patient is classified as non-diabetic. In medical screening, this is especially important because a missed case may delay further testing or treatment.

- Use SMOTE or class weights to increase the importance of diabetic cases.
- Adjust the decision threshold to favour higher recall.
- Use recall and F1-Score as important evaluation measures instead of accuracy alone.
- Use cross-validation to ensure the improvement is consistent.

Clinical justification: A screening model should minimise missed diabetes cases because identifying a potentially diabetic patient for further clinical assessment is generally safer than relying only on a high overall accuracy.

TASK 2 (CONTINUED): PRUNING & TASK 3

(c) Post-pruning using cost-complexity pruning

Post-pruning starts with a trained Decision Tree and removes branches that contribute little to generalisation. Cost-complexity pruning balances prediction error against tree size:

$R_{\alpha}(T) = R(T) + \alpha|T|$

Here, R(T) represents the error of the tree, |T| represents the number of terminal nodes, and α controls the penalty for complexity. Increasing α encourages a smaller tree.

Feature	Pre-pruning	Post-pruning
When applied	During tree construction	After a larger tree has been built
Main idea	Limit depth, minimum samples or other growth parameters	Remove branches that do not improve generalisation
Advantage	Can reduce overfitting early	Can use information from the fully grown tree to remove weak branches
Clinical value	Produces simpler trees	Can produce a compact and interpretable final tree

Deployment choice: The better model should be selected using validation/test performance, especially diabetic-class recall and F1-Score, while also considering interpretability and stability. A post-pruned model is attractive for clinical deployment when it achieves comparable or better performance with fewer unnecessary branches.

The paper does not provide actual model outputs, so exact pre-pruning and post-pruning accuracy values cannot be calculated from the question sheet alone.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Logistic Regression

Logistic Regression is a suitable statistical learning model for binary diabetes prediction. It estimates the probability of the positive class using the logistic function:

$P(Y=1) = 1 / (1 + e^{-z})$

where z is a weighted combination of the patient features. A probability threshold can then be used to classify the patient as diabetic or non-diabetic.

Aspect	Decision Tree	Logistic Regression
Relationship	Can capture non-linear thresholds and interactions.	Models a statistical relationship through weighted predictors.
Interpretability	Produces simple if-then rules.	Coefficients show direction and strength of association.
Probability	Can provide class probabilities.	Naturally provides probability estimates.
Scaling	Usually not essential.	Usually beneficial for numerical predictors.
Best use	Rule-based clinical decision support.	Risk scoring and probability-based stratification.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

Performance comparison: Accuracy and AUC-ROC

The Decision Tree and Logistic Regression should be evaluated on the same test set. **Accuracy** measures the overall proportion of correct predictions. **AUC-ROC** measures how well the model separates diabetic and non-diabetic patients over different classification thresholds. A higher AUC indicates stronger discrimination.

A statistical model is preferable when the application needs a compact mathematical model, probability/risk estimates, and a relatively direct interpretation of how predictors influence the output. A Decision Tree is preferable when rule-based explanations and non-linear thresholds are important.

Numerical limitation: The source assessment specifies Accuracy and AUC-ROC but does not contain patient records or trained-model results. Therefore, exact numerical Accuracy and AUC values cannot be invented.

(b) Feature importance analysis

Feature importance should be performed separately for the Decision Tree and the statistical model. For the Decision Tree, importance is commonly based on the total reduction in impurity contributed by a feature. For Logistic Regression, the magnitude of standardised coefficients can be compared to understand relative influence.

Three important predictors to investigate:

- **1. Glucose level** – likely to be a strong predictor because glucose directly relates to diabetes risk.
- **2. BMI** – provides information about body composition and associated metabolic risk.
- **3. Age / Blood pressure** – both can provide additional risk information; the actual third-ranked feature must be determined from model output.

Do the models agree? They may agree on the strongest predictors, but they do not have to produce identical rankings. The Decision Tree looks for features that create useful splits and can capture interactions and thresholds. Logistic Regression estimates predictor effects within its statistical form. Therefore, differences in feature ranking can be explained by the different learning mechanisms.

Interpretation

If glucose is ranked highly by both models, it provides stronger evidence that it is consistently useful for prediction. If one model ranks BMI highly while the other ranks age or blood pressure higher, the difference should be investigated using validation performance and domain knowledge rather than assuming one model is automatically correct.

TASK 4: REINFORCEMENT LEARNING FOR ACTION RECOMMENDATION

(a) Treatment recommendation as a Markov Decision Process

The treatment recommendation problem can be represented as an MDP because the agent observes a patient's health state, selects an action, receives a reward based on the outcome, and observes a new state.

TASK 4: REINFORCEMENT LEARNING FOR ACTION RECOMMENDATION

MDP component	Definition	Justification
States	Stable, At-Risk, High-Risk, Improving	Represent different stages of patient health.
Actions	Diet, Exercise, Medication, Monitor	These are the treatment/support actions specified in the question.
Reward	Health improvement score; positive for improvement and negative for deterioration.	Encourage actions that improve patient outcomes.
Transition	Current patient state + chosen action → next patient state.	Represents how health condition changes over time.

(b) Q-Learning for at least 5 patient episodes

Q-Learning learns the value of each state-action pair without requiring a complete transition model. The update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

For a worked illustration, assume learning rate $\alpha = 0.5$ and discount factor $\gamma = 0.9$. Since the source does not provide actual patient transitions or rewards, the following values demonstrate the requested update procedure rather than claiming measured clinical results.

Episode / Iteration	State	Action	Reward	Updated Q-value
1 / I1	At-Risk	Diet	+2	1.000
1 / I1	High-Risk	Monitor	+1	0.500
2 / I1	At-Risk	Exercise	+3	1.500
3 / I2	At-Risk	Diet	+2	1.675
4 / I2	High-Risk	Medication	+4	2.675
5 / I2	Improving	Monitor	+2	1.000

Example calculation: For At-Risk → Diet in Iteration 2, using previous $Q = 1.00$ and next-state maximum $Q = 1.50$:

$$Q = 1.00 + 0.5[2 + 0.9(1.50) - 1.00] = 1.675$$

Convergence criterion: Training can be stopped when the maximum absolute change in Q-values becomes less than 0.01 for several consecutive episodes, or after a predefined maximum number of episodes.

(c) Final learned policy

Patient state	Illustrative learned action
Stable	Monitor
At-Risk	Diet / Exercise
High-Risk	Medication / Exercise*
Improving	Monitor

Ethical considerations

- **Patient safety:** The RL agent must not independently prescribe or change medication without qualified medical supervision.
- **Bias:** Training data should represent different patient groups fairly; biased data can produce biased recommendations.
- **Explainability:** Clinicians should be able to understand the reason for a recommendation and challenge it when necessary.
- **Privacy:** Patient records must be securely stored and used only for authorised healthcare purposes.
- **Accountability:** A qualified healthcare professional should retain responsibility for the final treatment decision.
- **Safe reward design:** Rewards must discourage unnecessary medication or interventions and prioritise patient well-being.

*Medication decisions must remain under qualified clinical supervision.

Conclusion: The proposed system combines inductive learning for diabetes diagnosis, a Decision Tree for interpretable classification, statistical learning for risk probability and Q-Learning for sequential treatment-action recommendation. The

exact numerical model results must be calculated from the actual patient dataset; the worked Q-values above are illustrative calculations based on the specified Q-Learning method.